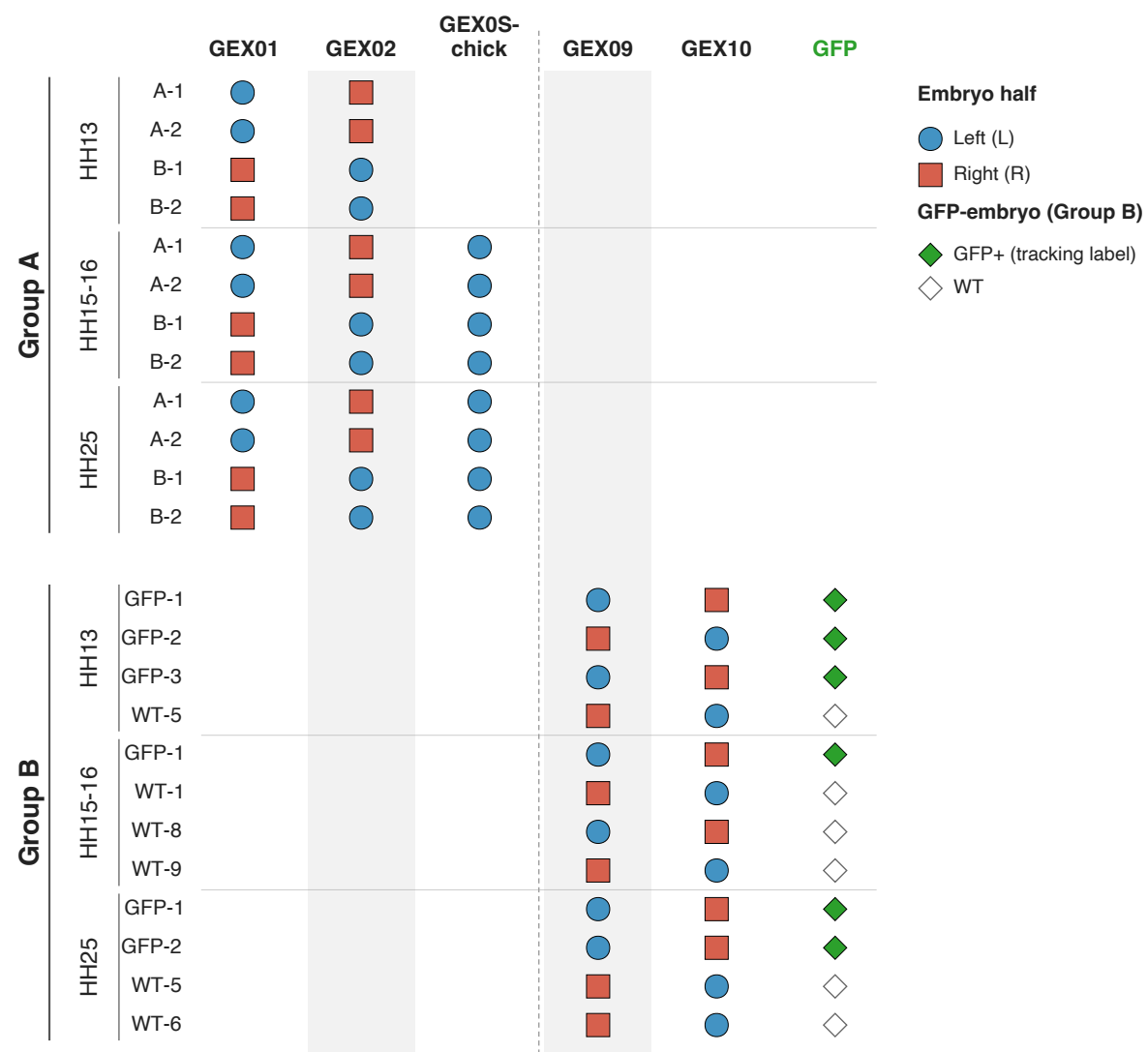
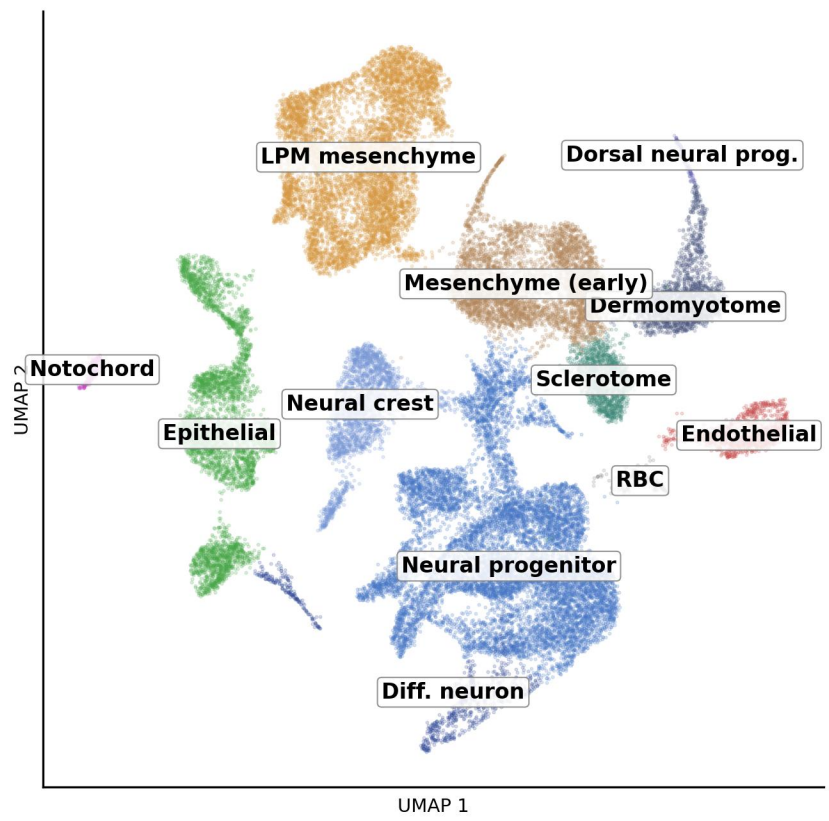
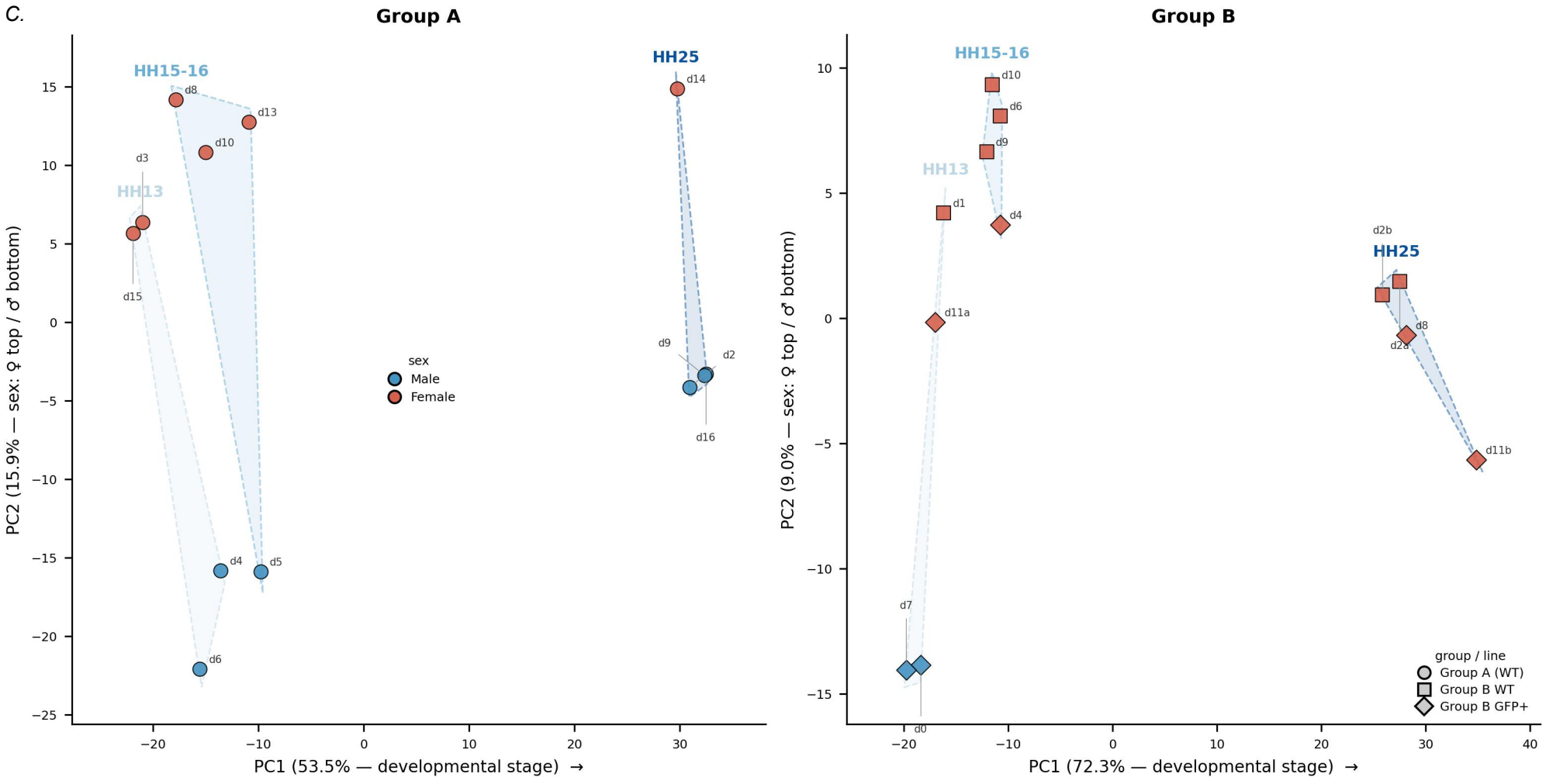
A.


GEX01
GEX02
GEX0S-chick
GEX09
GEX10
GFP
Group A
HH13
HH15-16
HH25
A-1
A-2
B-1
B-2
Group B
GFP-1
GFP-2
GFP-3
WT-5
WT-1
WT-8
WT-9
WT-6
Embryo half
Left (L)
Right (R)
GFP-embryo (Group B)
GFP+ (tracking label)
WT

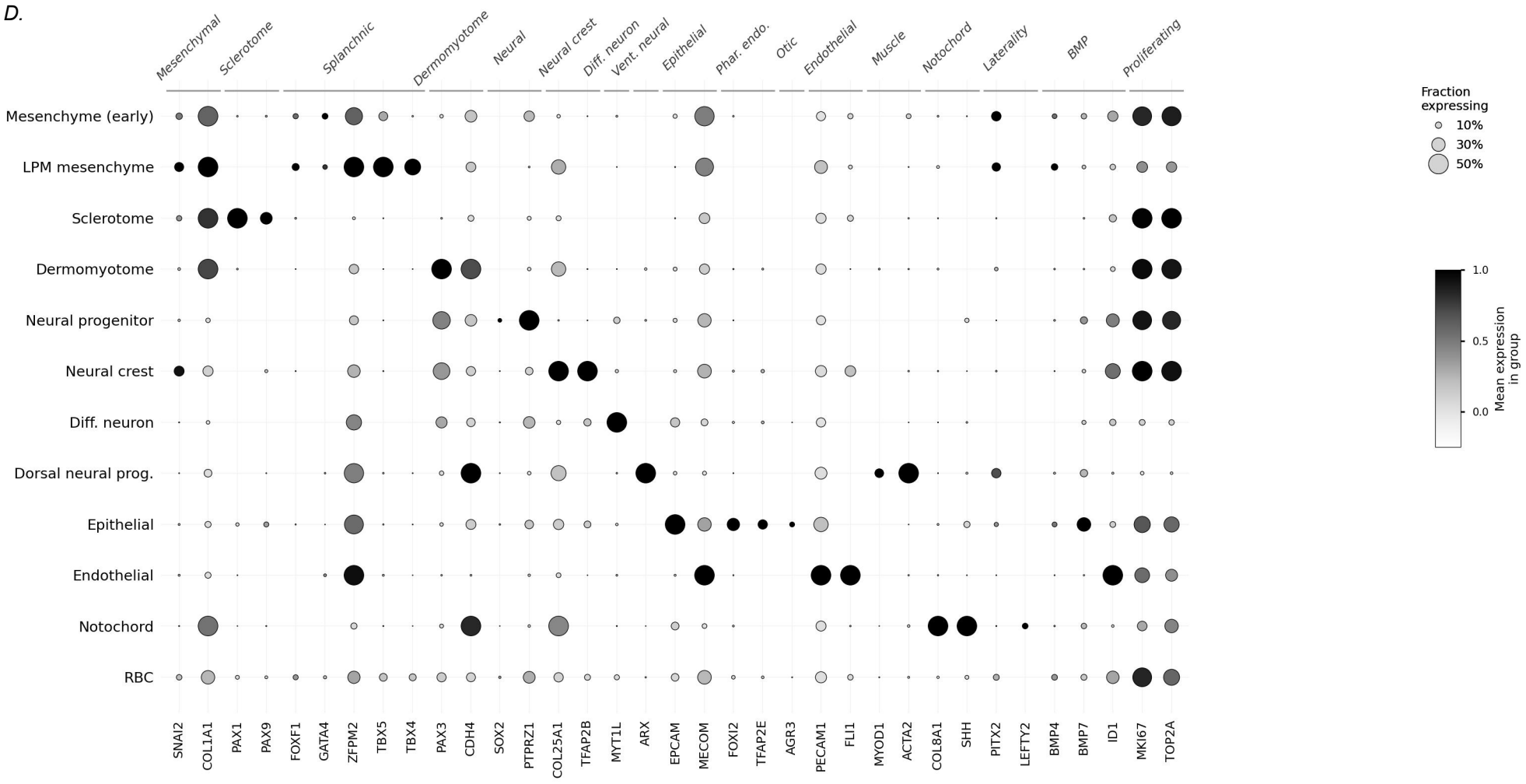
B.


LPM mesenchyme
Dorsal neural prog.
Mesenchyme (early)
Dermomyotome
Notochord
Sclerotome
Neural crest
Epithelial
Endothelial
RBC
Neural progenitor
Diff. neuron
UMAP 2
UMAP 1

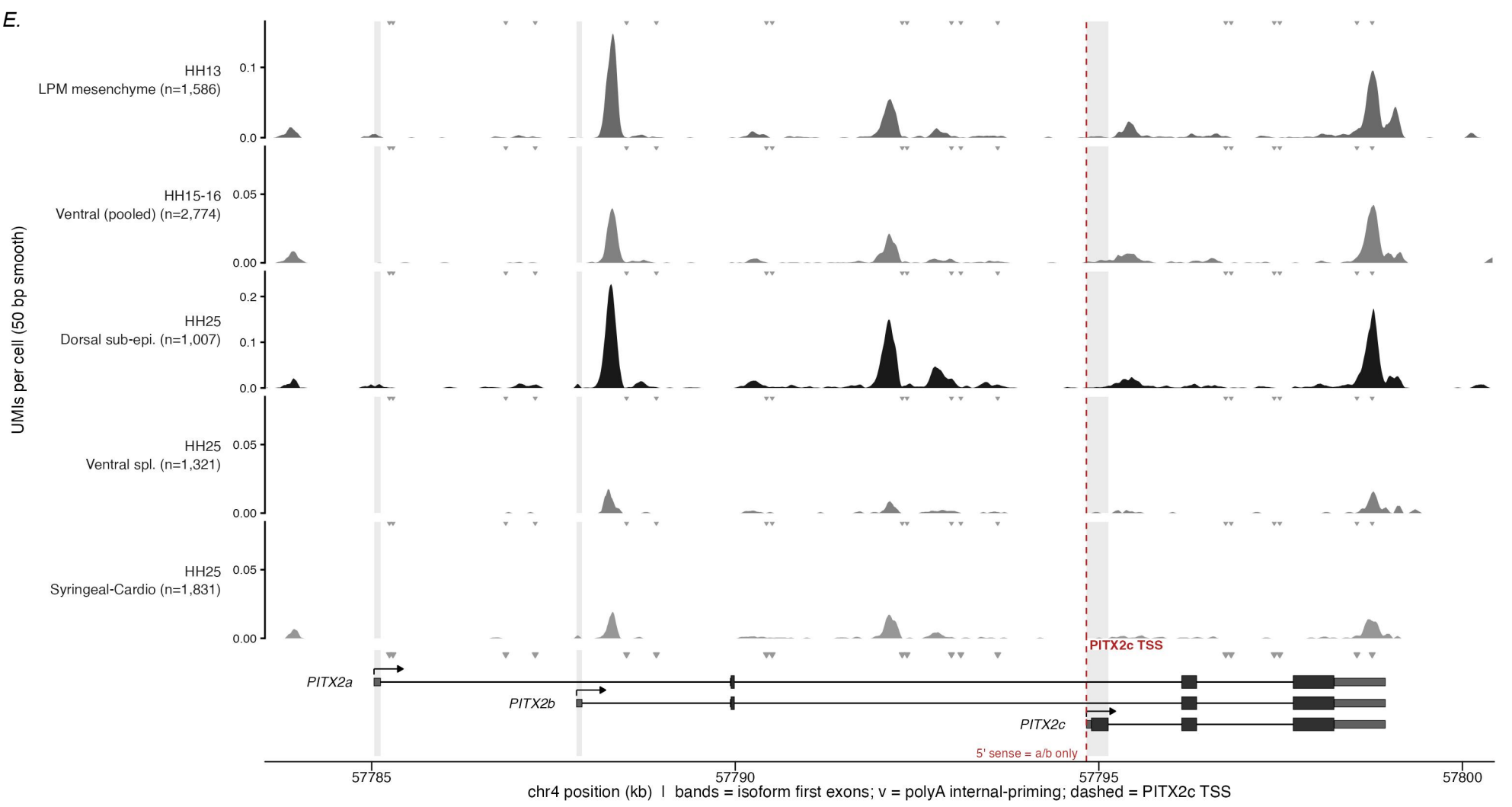
C.


Group A
Group B
HH15-16
HH13
HH25
sex
Male
Female
group / line
Group A (WT)
Group B WT
Group B GFP+
PC2 (15.9% — sex: ♀ top / ♂ bottom)
PC2 (9.0% — sex: ♀ top / ♂ bottom)
PC1 (53.5% — developmental stage) →
PC1 (72.3% — developmental stage) →

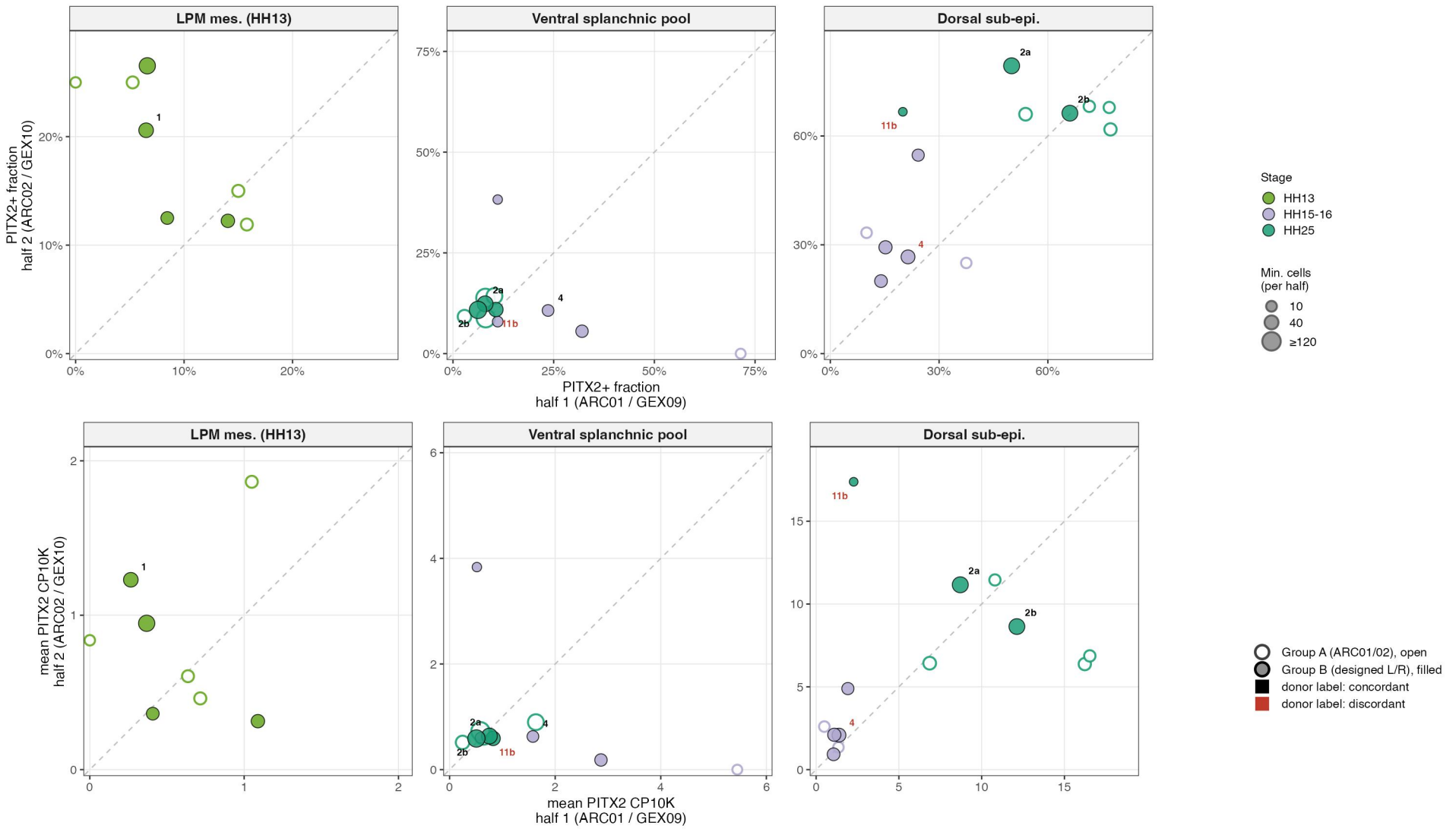
D.


Mesenchymal
Sclerotome
Splanchnic
Dermomyotome
Neural
Neural crest
Diff. neuron
Vent. neural
Epithelial
Phar. endo.
Otic
Endothelial
Muscle
Notochord
Laterality
BMP
Proliferating
Mesenchyme (early)
LPM mesenchyme
Sclerotome
Dermomyotome
Neural progenitor
Neural crest
Diff. neuron
Dorsal neural prog.
Epithelial
Endothelial
Notochord
RBC
SNAI2
COL1A1
PAX1
PAX9
FOXF1
GATA4
ZFPM2
TBX5
TBX4
PAX3
CDH4
SOX2
PTPRZ1
COL25A1
TFAP2B
MYT1L
ARX
EPCAM
MECOM
FOXI2
TFAP2E
AGR3
PECAM1
FLI1
MYOD1
ACTA2
COL8A1
SHH
PITX2
LEFTY2
BMP4
BMP7
ID1
MKI67
TOP2A
Fraction expressing
10%
30%
50%
Mean expression in group

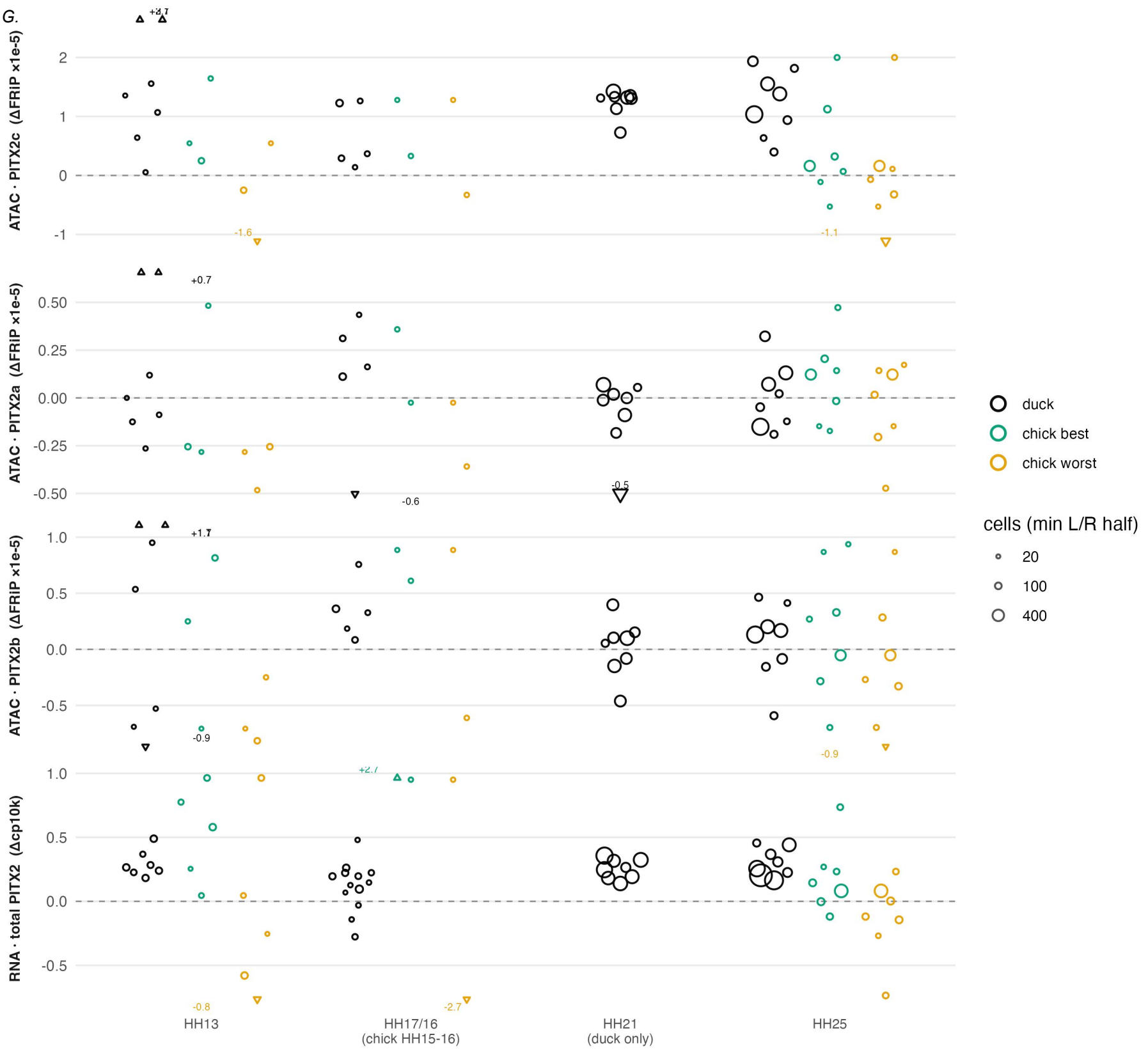
E.


HH13
LPM mesenchyme (n=1,586)
HH15-16
Ventral (pooled) (n=2,774)
HH25
Dorsal sub-epi. (n=1,007)
HH25
Ventral spl. (n=1,321)
HH25
Syringeal-Cardio (n=1,831)
UMIs per cell (50 bp smooth)
PITX2c TSS
PITX2a
PITX2b
PITX2c
5' sense = a/b only
57785
57790
57795
57800
chr4 position (kb) | bands = isoform first exons; v = polyA internal-priming; dashed = PITX2c TSS

F.


LPM mes. (HH13)
Ventral splanchnic pool
Dorsal sub-epi.
PITX2+ fraction half 2 (ARC02 / GEX10)
PITX2+ fraction half 1 (ARC01 / GEX09)
mean PITX2 CP10K half 2 (ARC02 / GEX10)
mean PITX2 CP10K half 1 (ARC01 / GEX09)
Stage
HH13
HH15-16
HH25
Min. cells (per half)
10
40
≥120
Group A (ARC01/02), open
Group B (designed L/R), filled
donor label: concordant
donor label: discordant

G.


ATAC · PITX2c (ΔFRiP ×1e-5)
ATAC · PITX2a (ΔFRiP ×1e-5)
ATAC · PITX2b (ΔFRiP ×1e-5)
RNA · total PITX2 (Δcp10k)
HH13
HH17/16 (chick HH15-16)
HH21 (duck only)
HH25
duck
chick best
chick worst
cells (min L/R half)
20
100
400